

Assignment 5

Name: _____

Student ID: _____

12 problems. Show all of your work.

Problem 1. True or False

If $\vec{u}$ and $\vec{v}$ are two non-zero vectors in $\mathbb{R}^3$, then $\vec{u} \cdot (\vec{u} \times \vec{v}) = \|\vec{u}\|^2 + \vec{u} \cdot \vec{v}$.

Justify your answer fully: give a proof if the statement is true, or a counterexample if it is false.

Problem 2.

Let $\mathbf{v}$ be the vector $3\mathbf{i} - 4\mathbf{j} + 5\mathbf{k}$.

- A. Find a non-zero vector $\mathbf{v}'$ orthogonal to $\mathbf{v}$.
- B. Find a vector orthogonal to both $\mathbf{v}$ and $\mathbf{v}'$.

Problem 3.

Suppose $\|\vec{u}\| = 3$, $\|\vec{v}\| = 4$ and $\|\vec{u} \times \vec{v}\| = 6$. Find

- (a) $|\vec{u} \cdot \vec{v}|$;
- (b) $\|\text{proj}_{\vec{u}} \vec{v}\|$.

Problem 4.

Find the Cartesian equation $f(x, y) = 0$ for the curve represented by the following parametric equations, then graph it:

$$\begin{aligned}x(t) &= e^{2t}, \\y(t) &= e^t - 1, \\0 &\leq t \leq \ln(2).\end{aligned}$$

Problem 5.

Provide a parametrized form of the curve given by

$$x^2 + 4y^2 + 8y - 2\sqrt{21}x = 0.$$

Assume that the curve is traced **counter-clockwise** as the parameter increases.

Problem 6.

- A. Parametrize the intersection of $x^2 + y^2 + z^2 = 25$ and the plane $x = 4$.
B. Which axis is perpendicular to the plane $x = 4$?

Problem 7.

Consider the curve $25(y + 4)^2 + 400(x - 1)^2 = 100$.

- (a) Sketch the curve and label its important points.
(b) Parametrize the entire curve using one parametrization.

Problem 8.

Find the vector equation of the line of intersection of the following planes:

$$\begin{aligned}2x + y - z &= 6, \\ x - y + 3z &= 4.\end{aligned}$$

Problem 9.

Find an equation of a plane containing the line $\mathbf{r} = \langle -4, 5, -1 \rangle + t\langle -4, 6, -3 \rangle$ and parallel to the plane $-3x - y + 2z = 0$, in which the coefficient of x is -3 .

Problem 10.

Calculate the distance between the following two lines:

$$\begin{aligned}L_1 : \quad x &= 4 - t, \quad y = 3 + 2t, \quad z = -5 + 3t \\ L_2 : \quad x &= 5 - t, \quad y = 3 + 2t, \quad z = -5 + 3t\end{aligned}$$

Problem 11.

Find parametric equations for the line through $(2, 1, 1)$ which is normal to the plane containing $(0, 1, 1)$, $(-1, 2, 4)$ and $(3, 0, 1)$.

Problem 12.

Let L be the line given by

$$x = 4 + t, \quad y = -1 - 2t, \quad z = 5 + t.$$

Find the point on L that is closest to $(-2, 2, -1)$.